

MAT 362 HW02

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Problem. 1.5.1. Given the parametrized curve

$$\alpha(s) = \left(a \cos \frac{s}{c}, a \sin \frac{s}{c}, b \frac{s}{c} \right), \quad s \in \mathbb{R},$$

where $c^2 = a^2 + b^2$,

- (a) Show that the parameter s is the arc length.
- (b) Determine the curvature and torsion of α .
- (c) Determine the osculating plane of α .
- (d) Show that the lines containing $n(s)$ and passing through $\alpha(s)$ meet the z -axis under a constant angle equal to $\pi/2$.
- (e) Show that the tangent lines to α make a constant angle with the z -axis.

Solution. (a) It suffices to show that $|\alpha'(s)| = 1$. To see this, we compute

$$|\alpha'(s)| = \sqrt{\left(-\frac{a}{c} \sin \frac{s}{c}\right)^2 + \left(\frac{a}{c} \cos \frac{s}{c}\right)^2 + \left(\frac{b}{c}\right)^2} = \sqrt{\frac{a^2 + b^2}{c^2}} = 1.$$

- (b) The curvature of α is equal to $|\alpha''(s)|$, which in this case is

$$\left| \left(-\frac{a}{c^2} \cos \frac{s}{c}, -\frac{a}{c^2} \sin \frac{s}{c}, 0 \right) \right| = \boxed{\frac{a}{c^2}}.$$

The normal vector $n(s)$ is then $(-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0)$ and we know that $t(s) = (-\frac{a}{c} \sin \frac{s}{c}, \frac{a}{c} \cos \frac{s}{c}, \frac{b}{c})$, so their vector product is

$$b(s) = t(s) \wedge n(s) = \left(\frac{b}{c} \sin \frac{s}{c}, -\frac{b}{c} \cos \frac{s}{c}, \frac{a}{c} \right),$$

so the torsion of α is

$$|b'(s)| = \left| \left(\frac{b}{c^2} \cos \frac{s}{c}, \frac{b}{c^2} \sin \frac{s}{c}, 0 \right) \right| = \boxed{\frac{b}{c^2}}.$$

- (c) The osculating plane of α is the plane through $\alpha(s)$ determined by $t(s)$ and $n(s)$, i.e. orthogonal to $b(s)$. This means it is the set of points x such that

$$(x - \alpha(s)) \cdot b(s) = 0.$$

Expanding this yields the plane

$$\boxed{x_1 \cdot b \sin \frac{s}{c} - x_2 \cdot b \cos \frac{s}{c} + ax_3 = abs}.$$

(d) We computed above that $n(s) = (-\cos \frac{s}{c}, -\sin \frac{s}{c}, 0)$, so the line in the direction of $n(s)$ passing through $\alpha(s)$ is

$$\ell(x) = \alpha(s) + x \cdot n(s)$$

for $x \in \mathbb{R}$. This line meets the z -axis at $x = a$, which is the point

$$\ell(a) = \left(0, 0, \frac{bs}{c}\right).$$

It is clear that the line makes a right angle with the z -axis because $n(s) \cdot (0, 0, 1) = 0$ means that $n(s)$ and the unit vector in the direction of the z -axis are orthogonal.

(e) The tangent lines to α make an angle of

$$\arccos(t(s) \cdot (0, 0, 1)) = \arccos\left(\left(-\frac{a}{c} \sin \frac{s}{c}, \frac{a}{c} \cos \frac{s}{c}, \frac{b}{c}\right) \cdot (0, 0, 1)\right) = \arccos \frac{b}{c}$$

with the z -axis, which is independent of our only parameter s . $\square$

Problem. 1.5.6. A **translation** by a vector $v \in \mathbb{R}^3$ is the map $A: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that is given by $A(p) = p + v$. A linear map $\rho: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is an **orthogonal transformation** if $\rho u \cdot \rho v = u \cdot v$ for all $u, v \in \mathbb{R}^3$. A **rigid motion** in $\mathbb{R}^3$ is the result of composing a translation with an orthogonal transformation with positive determinant.

- (a) Demonstrate that the norm of a vector and the angle θ between two vectors, $0 \leq \theta \leq \pi$, are invariant under orthogonal transformations with positive determinant.
- (b) Show that the vector product of two vectors is invariant under orthogonal transformations with positive determinant. Is the assertion still true if we drop the condition on the determinant?
- (c) Show that the arc length, the curvature, and the torsion of a parametrized curve are (whenever defined) invariant under rigid motions.

Solution. (a) Let ρ be an orthogonal transformation and let $u \in \mathbb{R}^3$. Then

$$\|u\|^2 = u \cdot u = \rho u \cdot \rho u = \|\rho u\|^2$$

implies norm is preserved under orthogonal transformations. Moreover, if θ is the angle between two vectors $u, v \in \mathbb{R}^3$,

$$\cos \theta = \frac{u \cdot v}{\|u\| \|v\|} = \frac{\rho u \cdot \rho v}{\|\rho u\| \|\rho v\|} = \cos \theta'$$

where θ' is the angle between ρu and ρv (with $0 \leq \theta, \theta' \leq \pi$), which implies that $\theta = \theta'$ because $\cos$ is injective on $[0, \pi]$, i.e. angle is preserved by ρ .

(b) Let $u, v \in \mathbb{R}^3$ be two vectors. The above problem shows that ρu , ρv , and $\rho(u \wedge v)$ have the same norms as u , v , $u \wedge v$, and moreover, $\rho(u \wedge v)$ is orthogonal to ρu and ρv . This narrows down $\rho(u \wedge v)$ to two possible vectors: either $\rho u \wedge \rho v$ or $-\rho u \wedge \rho v$. This is where we can use the fact that ρ has positive determinant: since ρ preserves orientation, ρu , ρv , and $\rho(u \wedge v)$ have the same orientation as u , v , and $u \wedge v$, which confirms that $\rho(u \wedge v) = \rho u \wedge \rho v$.

Indeed, if ρ is allowed to have negative orientation, then consider the transformation given by the matrix

$$\rho = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

in the standard basis, i.e. reflection through the xy -plane. This is an orthogonal transformation because for any two $u, v \in \mathbb{R}^3$,

$$\rho u \cdot \rho v = (u_1, u_2, -u_3) \cdot (v_1, v_2, -v_3) = u_1 v_1 + u_2 v_2 + u_3 v_3 = u \cdot v.$$

However, if e_1, e_2, e_3 are the standard basis vectors, then

$$e_3 = e_1 \wedge e_2 = \rho e_1 \wedge \rho e_2 = -\rho e_3$$

because $\rho e_1 = e_1$ and $\rho e_2 = e_2$, so ρ doesn't preserve vector products.

(c) Let $\alpha(s)$ be an arc length parametrized curve. Let ρ be an orthogonal transformation with matrix

$$\rho = \begin{pmatrix} - & v_1 & - \\ - & v_2 & - \\ - & v_3 & - \end{pmatrix}$$

where v_1, v_2, v_3 are row vectors. To start with we show that tangent vectors are invariant under ρ :

$$(\rho\alpha)'(s) = \begin{pmatrix} v_1 \cdot \alpha(s) \\ v_2 \cdot \alpha(s) \\ v_3 \cdot \alpha(s) \end{pmatrix}' = \begin{pmatrix} v_1 \cdot \alpha'(s) \\ v_2 \cdot \alpha'(s) \\ v_3 \cdot \alpha'(s) \end{pmatrix} = \rho \cdot \alpha'(s).$$

In particular, the arc length of the curve $\rho\alpha(s)$ from $s = a$ to $s = b$ is

$$\int_a^b |\rho\alpha'(s)| ds = \int_a^b |\alpha'(s)| ds,$$

recalling that ρ preserves norm, so arc length is preserved by ρ . Then, the curvature is

$$|(\rho\alpha)''(s)| = |\rho \cdot \alpha''(s)| = |\alpha''(s)| = \kappa(s),$$

so it is also preserved by ρ . For the torsion, we can check first that the binormal of α at point s is mapped to the binormal of $\rho\alpha$ at s : clearly $t(s) = \alpha'(s)$ and $n(s) = \alpha''(s)/\kappa(s)$ are preserved by ρ , and part (b) says that their vector product is also preserved by ρ . Then we have that the torsion of $\rho\alpha$ is the norm of $(\rho b)'(s)$, which is again just $|b'(s)| = \tau(s)$, so finally torsion is preserved by ρ . $\square$

Problem. 1.5.8(a). The trace of the parametrized curve (arbitrary parameter)

$$\alpha(t) = (t, \cosh t), \quad t \in \mathbb{R}$$

is called the **catenary**. Show that the signed curvature of the catenary is $\kappa(t) = \frac{1}{\cosh^2 t}$.

Solution. Recall that $\cosh t = \frac{1}{2}(e^t + e^{-t})$ and $\sinh t = \frac{1}{2}(e^t - e^{-t})$. We can compute

$$\alpha'(t) = (1, \sinh t), \quad |\alpha'(t)| = \sqrt{1 + \sinh^2 t} = \cosh t$$

(note that we do not need to say $|\cosh t|$ because e^t and e^{-t} are always positive). Therefore the normalized tangent vector is

$$T(t) = \left(\frac{1}{\cosh t}, \tanh t \right).$$

We then compute

$$\frac{d}{ds} T(t) = \frac{dt}{ds} \cdot T'(t).$$

We know that

$$\frac{dt}{ds} = \frac{1}{ds/dt} = \frac{1}{|\alpha'(t)|},$$

so the derivative of T with respect to s is

$$\frac{T'(t)}{|\alpha'(t)|} = \frac{1}{\cosh t} \cdot \left(-\frac{\sinh t}{\cosh^2 t}, 1 - \frac{\sin^2 t}{\cosh^2 t} \right) = \frac{1}{\cosh^2 t} \left(-\tanh t, \frac{1}{\cosh t} \right),$$

noting that $\cosh^2 t - \sinh^2 t = 1$. On the other hand, the unit normal vector n such that $\{T, n\}$ has the same orientation as $\{e_1, e_2\}$ is the $\frac{\pi}{2}$ -counterclockwise rotation of T , which is

$$\left(-\tanh t, \frac{1}{\cosh t} \right).$$

By definition, the signed curvature is the scalar κ such that $T'(s) = \kappa n$, so we find that $\kappa = \frac{1}{\cosh^2 t}$ as desired. $\square$

Problem. 1.5.10. Consider the map

$$\alpha(t) = \begin{cases} (t, 0, e^{-1/t^2}), & t > 0 \\ (t, e^{-1/t^2}, 0), & t < 0 \\ (0, 0, 0) & t = 0 \end{cases}$$

- Prove that α is a differentiable curve.
- Prove that α is regular for all t and that the curvature $\kappa(t) \neq 0$ for $t \neq 0$, $t \neq \pm\sqrt{2/3}$, and $\kappa(0) = 0$.
- Show that the limit of the osculating planes as $t \rightarrow 0$, $t > 0$ is the plane $y = 0$ but that the limit of the osculating planes as $t \rightarrow 0$, $t < 0$, is the plane $z = 0$.
- Show that τ can be defined so that $\tau \equiv 0$, even though α is not a plane curve.

Solution. (a) Differentiability away from zero is trivial, so we just check that $\alpha'(0)$ exists. Specifically, we need to check that the functions

$$y(t) = \begin{cases} e^{-1/t^2} & t < 0 \\ 0 & t \geq 0 \end{cases} \quad z(t) = \begin{cases} e^{-1/t^2} & t > 0 \\ 0 & t \leq 0 \end{cases}$$

are differentiable at zero. Notice that $y(t) = z(-t)$, so we will verify differentiability for y only, since differentiability at 0 for one function implies the same for the other. This amounts to computing

$$\lim_{h \rightarrow 0} \frac{y(h) - y(0)}{h}.$$

From the left, this limit is clearly 0, so we need to show that

$$\lim_{h \rightarrow 0^+} \frac{e^{-1/h^2}}{h} = 0.$$

We have

$$0 \leq \lim_{h \rightarrow 0^+} \frac{e^{-1/h^2}}{h} = \lim_{x \rightarrow \infty} \frac{x}{e^{x^2}} \leq \lim_{x \rightarrow \infty} \frac{x}{x^2} = 0$$

where the latter inequality comes from $e^r > r^2$ for large enough r , so this confirms that $y'(0) = z'(0) = 0$.

(b) α is regular for all t because the x -component of α' is always 1, so $|\alpha'| \geq 1$ for all t . To compute the curvature we use the following fact, noting that α is essentially a plane curve when restricting to $t > 0$ or $t < 0$.

Claim (Problem 1.5.12(d)) — If $\alpha: I \rightarrow \mathbb{R}^2$ is a plane curve $\alpha(t) = (x(t), y(t))$, the signed curvature of α at t is

$$\kappa(t) = \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}}.$$

Proof. To start with we can calculate the unit tangent

$$T(t) = \left(\frac{x'}{\sqrt{(x')^2 + (y')^2}}, \frac{y'}{\sqrt{(x')^2 + (y')^2}} \right).$$

Then, the derivative of the unit tangent with respect to t is

$$\left(\frac{-y'(x'y'' - x''y')}{((x')^2 + (y')^2)^{3/2}}, \frac{x'(x'y'' - x''y')}{((x')^2 + (y')^2)^{3/2}} \right),$$

which can be verified by some menial computation. To get the derivative with respect to s , we just multiply this by $dt/ds = \frac{1}{|\alpha'(t)|}$. This means that, if n is the normal vector obtained by a $\frac{\pi}{2}$ -counterclockwise rotation of T , then the signed curvature is the scalar κ such that $T = \kappa n$: in this case

$$n = \left(\frac{-y'}{|\alpha'|}, \frac{x'}{|\alpha'|} \right),$$

and we know that

$$\frac{dT}{ds} = \frac{1}{|\alpha'|} \cdot \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}} \cdot (-y', x') = \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}} \cdot \frac{(-y', x')}{|\alpha'|},$$

so we can see that the curvature is indeed the expression in the claim. $\square$

We focus on $t \geq 0$, since $t < 0$ and $t > 0$ are symmetrical and so the computations for a given value of t end up being the same as for $-t$. For $t > 0$, the curvature is equal to

$$\frac{|x'z'' - x''z'|}{|\alpha'|^3} = \frac{|z''|}{|\alpha'|^3},$$

so we need to find $t > 0$ for which $z'' = 0$ in order to check when the curvature is 0. We have

$$z''(t) = \frac{-6}{t^4}e^{-1/t^2} + \frac{4}{t^6}e^{-1/t^2},$$

which is zero if and only if

$$6t^2 = 4,$$

so for $t > 0$ the only point of zero curvature is $t = \sqrt{2/3}$. Symmetrically, for $t < 0$, the only point of zero curvature is $t = -\sqrt{2/3}$.

Finally, for $t = 0$, we can tediously compute that the normal vector is the zero vector, which is enough to imply that the curvature is 0.

(c) For $t > 0$, the y -coordinate is always 0, so the curve as well as its tangent and normal vectors live in the $y = 0$ plane, hence the osculating plane is the $y = 0$ plane as well; similarly, the curve and its tangent and normal vectors all lie in the $z = 0$ plane for $t < 0$, so the osculating plane there is $z = 0$. The corresponding limits follow immediately.

(d) The torsions for $t > 0$ and for $t < 0$ are both zero because α lies in a single plane for each of $t > 0$ and $t < 0$. Moreover we can continuously extend τ to be 0 at $t = 0$, so we can reasonably have $\tau \equiv 0$ despite α not being a planar curve. $\square$

Problem. 1.5.11. One often gives a plane curve in polar coordinates by $\rho = \rho(\theta)$, $a \leq \theta \leq b$.

(a) Show that the arc length is

$$\int_a^b \sqrt{\rho^2 + (\rho')^2} d\theta,$$

where the prime denotes the derivative relative to θ .

(b) Show that the curvature is

$$\kappa(\theta) = \frac{2(\rho')^2 - \rho\rho'' + \rho^2}{((\rho')^2 + \rho^2)^{3/2}}.$$

Solution. (a) The curve can be expressed as $\alpha(\theta) = (\rho(\theta) \cos \theta, \rho(\theta) \sin \theta)$. Then, its arc length is

$$\int_a^b |\alpha'(\theta)| d\theta.$$

We see that

$$\alpha'(\theta) = (-\rho \sin \theta + \rho' \cos \theta, \rho \cos \theta + \rho' \sin \theta),$$

so that we can calculate $|\alpha'(\theta)| = \sqrt{\rho^2 + (\rho')^2}$, justifying the given arc length formula.

(b) We use the following formula (proof copied over from the previous problem):

Claim (Problem 1.5.12(d)) — If $\alpha: I \rightarrow \mathbb{R}^2$ is a plane curve $\alpha(t) = (x(t), y(t))$, the signed curvature of α at t is

$$\kappa(t) = \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}}.$$

Proof. To start with we can calculate the unit tangent

$$T(t) = \left(\frac{x'}{\sqrt{(x')^2 + (y')^2}}, \frac{y'}{\sqrt{(x')^2 + (y')^2}} \right).$$

Then, the derivative of the unit tangent with respect to t is

$$\left(\frac{-y'(x'y'' - x''y')}{((x')^2 + (y')^2)^{3/2}}, \frac{x'(x'y'' - x''y')}{((x')^2 + (y')^2)^{3/2}} \right),$$

which can be verified by some menial computation. To get the derivative with respect to s , we just multiply this by $dt/ds = \frac{1}{|\alpha'(t)|}$. This means that, if n is the normal vector obtained by a $\frac{\pi}{2}$ -counterclockwise rotation of T , then the signed curvature is the scalar κ such that $T = \kappa n$: in this case

$$n = \left(\frac{-y'}{|\alpha'|}, \frac{x'}{|\alpha'|} \right),$$

and we know that

$$\frac{dT}{ds} = \frac{1}{|\alpha'|} \cdot \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}} \cdot (-y', x') = \frac{x'y'' - x''y'}{((x')^2 + (y')^2)^{3/2}} \cdot \frac{(-y', x')}{|\alpha'|},$$

so we can see that the curvature is indeed the expression in the claim. $\square$

We have already compute x' and y' above; we can also calculate

$$\begin{aligned}x'' &= -\rho \cos \theta - \rho' \sin \theta - \rho' \sin \theta + \rho'' \cos \theta = -\rho \cos \theta - 2\rho' \sin \theta + \rho'' \cos \theta \\y'' &= -\rho \sin \theta + \rho' \cos \theta + \rho' \cos \theta + \rho'' \sin \theta = -\rho \sin \theta + 2\rho' \cos \theta + \rho'' \sin \theta.\end{aligned}$$

Then

$$\begin{aligned}x'y'' &= (-\rho \sin \theta + \rho' \cos \theta)(-\rho \sin \theta + 2\rho' \cos \theta + \rho'' \sin \theta) \\&= \rho^2 \sin^2 \theta - 3\rho\rho' \cos \theta \sin \theta - \rho\rho'' \sin^2 \theta + 2(\rho')^2 \cos^2 \theta + \rho'\rho'' \cos \theta \sin \theta \\y'x'' &= (\rho \cos \theta + \rho' \sin \theta)(-\rho \cos \theta - 2\rho' \sin \theta + \rho'' \cos \theta) \\&= -\rho^2 \cos^2 \theta - 3\rho\rho' \cos \theta \sin \theta + \rho\rho'' \cos^2 \theta - 2(\rho')^2 \sin^2 \theta + \rho'\rho'' \cos \theta \sin \theta\end{aligned}$$

so the difference is

$$x'y'' - x''y' = \rho^2 - \rho\rho'' + 2(\rho')^2,$$

hence the signed curvature is

$$\frac{x'y'' - x''y'}{|\alpha'|^3} = \frac{\rho^2 - \rho\rho'' + 2(\rho')^2}{(\rho^2 + (\rho')^2)^{3/2}},$$

but this is equal to the unsigned curvature because the numerator is always nonnegative, for the expression $a^2 - ab + 2b^2$ is always nonnegative a, b as it's equal to a linear combination of squares:

$$\left(a - \frac{b}{2}\right)^2 + \frac{7b^2}{4}. \quad \square$$

Problem. 1.7.1. Is there a simple closed curve in the plane with length equal to 6 feet and bounding an area of 3 square feet?

Solution. No; the isoperimetric inequality says that

$$\ell^2 \geq 4\pi A$$

must hold, where ℓ is the length of the curve and A is the area bounded by the curve, but we can see clearly that

$$6^2 = 12 \cdot 3 < 4\pi \cdot 3 = 12\pi,$$

so this combination of area and arc length is impossible. $\square$

Problem. 1.7.2. Let $\overline{AB}$ be a segment of a straight line and let ℓ be greater than the length of $\overline{AB}$. The curve γ with endpoints A and B has length ℓ and maximizes the area bounded by γ and segment $\overline{AB}$. Show that γ is an arc of a circle passing through A and B .

Solution. We will take for granted that $\gamma: [0, 1] \rightarrow \mathbb{R}^2$ with $\gamma(0) = A$, $\gamma(1) = B$ is a parametrized non self-intersecting regular curve. In an abuse of notation, we will also refer to the image of γ by γ (so for example when we say a point P “lies on γ ” we mean that there is a $t \in [0, 1]$ such that $\gamma(t) = P$).

Also, let $\gamma': [0, 2] \rightarrow \mathbb{R}^2$ be the curve such that $\gamma'|_{[0, 1]} = \gamma$ and $\gamma'|_{[1, 2]}$ is a parametrization of segment $\overline{AB}$ with $\gamma'(1) = B$, $\gamma'(2) = \gamma'(0) = A$. Then γ' is a simple closed piecewise regular parametrized curve. Let R be the region whose area we are trying to maximize (i.e. the region bounded by γ'). We first show the following claim:

Claim — If line $\overline{AB}$ intersects the interior of R , then the area of R is not maximal.

Proof. Suppose line $\overline{AB}$ intersects the interior of R . and orient the plane so that $\overline{AB}$ lies in the x -axis with A to the left of B (so as t increases, the point $\gamma'(t)$ moves “clockwise” along the curve). Moreover, $\overline{AB}$ borders an “inside region” and an “outside region”; if necessary, reflect the plane about the x -axis so that the side of $\overline{AB}$ bordering the “outside region” is below the x -axis. That is, if an ant were to stand at an arbitrary point on segment $\overline{AB}$ (but not on A or B) and then walk ε units in the negative y direction for any sufficiently small $\varepsilon > 0$, it would be outside R .

Let $T(t)$ be the tangent line to γ at the point $\gamma(t)$. As t varies from 0 to 1, this line varies continuously. If $T(0)$ does not intersect the interior of R , then one can rotate this line counterclockwise until it first intersects γ at a point lying below the x -axis; let λ be the resulting line.

Otherwise, if $T(0)$ does intersect the interior of R , then consider the infimum t_0 of all t such that $T(t)$ does not intersect the interior of R (there exist such t because the tangent line at the point of γ with maximal y -coordinate is necessarily horizontal and thus can't intersect the interior of R). The line $T(t_0)$ intersects γ at $P = \gamma(t_0)$ and another point Q such that P, A, B , and Q lie on γ' in counterclockwise order. In this case let $\lambda = T(t_0)$.

In either case, we can reflect $\overline{AB}$ and the arcs joining A and B to the nearest points of γ lying on λ over the line λ . Since γ can only intersect the x -axis finitely many times except at arcs lying entirely within the x -axis (maybe more accurately phrased as “can only start intersecting the x -axis finitely many times”), we can repeat this process until eventually the resulting curve is entirely within one of the half-planes generated by the image of line $\overline{AB}$, and each reflection increases the area of the bounded region while preserving the arc length; thus we have constructed a $\hat{\gamma}$ that bounds a larger area than that of R . $\square$

Now let O be a point such that $OA = OB = r$ and O and C are on different sides of line AB . If $2\theta = \angle AOB$, then we impose the restriction

$$\frac{2\theta}{2\pi} = \frac{\ell}{2\pi r} \iff \ell = 2r\theta.$$

To see that such an (r, θ) pair exists, we can just compute directly that

$$AB = 2r \sin \theta = \frac{\ell \sin \theta}{\theta}.$$

This expression achieves 0 at $\theta = \pi/2$ and is equal to ℓ as θ approaches 0, meaning $\frac{\ell \sin \theta}{\theta}$ achieves every value between $[0, \ell)$ as θ varies over $[0, \pi/2]$; the fact that $AB < \ell$ then guarantees the existence of a solution by the intermediate value theorem.

This means that if C' is the arc centered at O with endpoints A and B lying on the opposite side of AB as C , then the closed curve γ composed of C and C' is a piecewise regular curve with fixed arc length $2\pi r$. Since the area of the region bounded by C' and segment AB is fixed, maximizing the area of R is equivalent to maximizing the area of the region bounded by γ . By the isoperimetric inequality, the area is maximized when γ is the circle centered at O with radius r . which exactly implies that C is an arc of a circle passing through A and B (an arc of γ to be precise). $\square$

(what is the nice way to do this problem? it caused me too much pain)